Introduction to the Technical Blueprint

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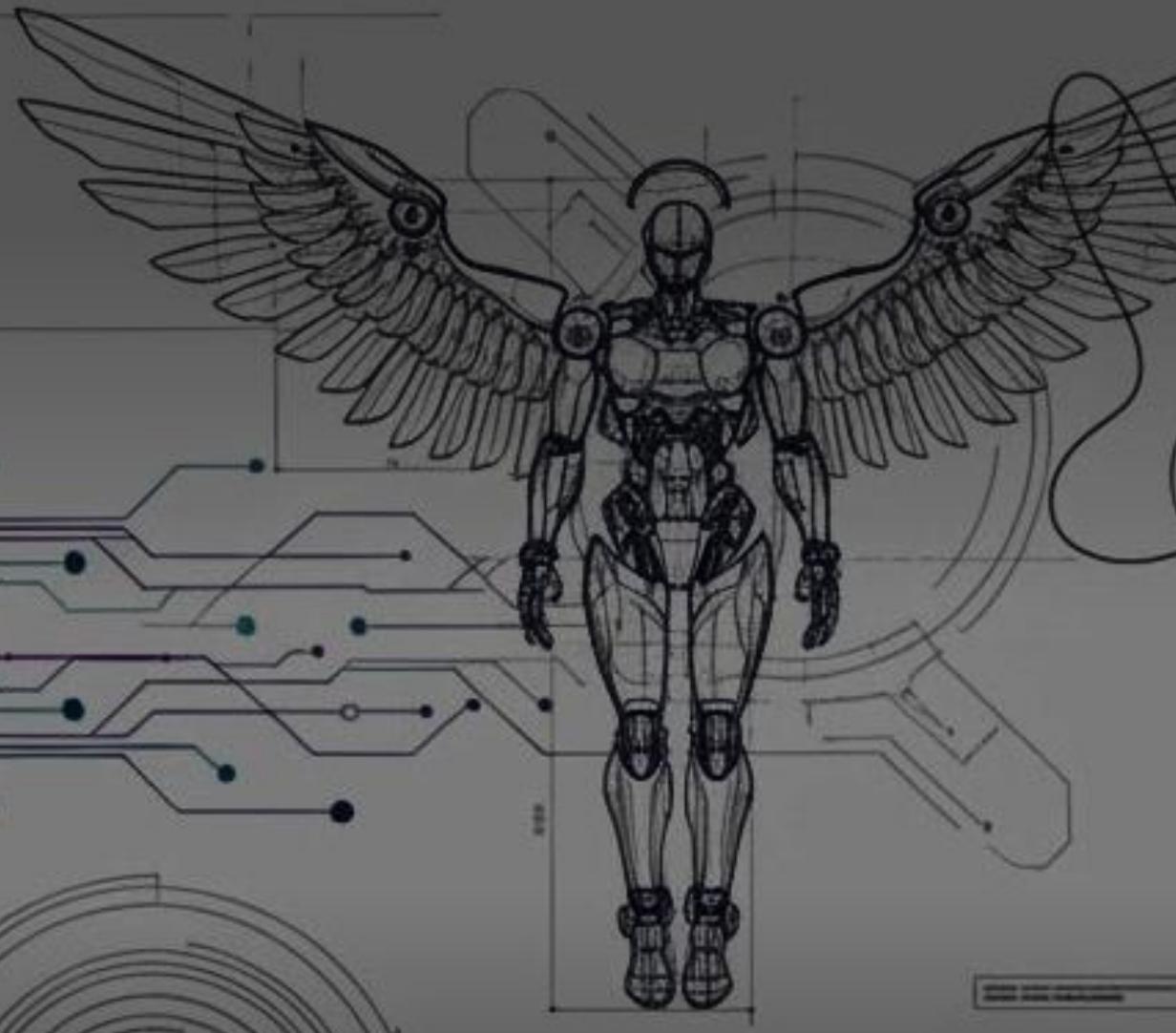

SERAPH'S DESPAIR

The Technical Blueprint: Seraph's Despair Goals

- Key Goal
  - The Technical Design Document (TDD) defines the ‘**how**’, focusing on stability, performance, and adherence to core design pillars.
- Core Technical Goals
  - **Performance:** Achieve 60 FPS on target platforms.
  - **Dual-State Seamlessness:** Support a rapid transition between **Platforming** (Real-time) and **Turn-Based Combat** states, minimizing "turn-based bloat".
  - **Scalability:** Ensure the Memory/Ability system can easily scale with new content.

Architectural Foundation: Godot & GDExtension

- Engine Stack
  - **Godot Engine** (Version TBD) for 2D Pixel Art rendering.
- GDExtension Strategy
  - **Separation of Concerns** to balance rapid development with performance.
- GDScript/C# (res://)
  - Used for high-level game logic, UI, and rapid prototyping.
- GDExtension (C++)
  - Used for **high-performance, complex systems** where native speed is critical:
    Soul Power Calculation,
    intensive *Memory Management*,
    and optimized *AI Logic*.

Project Layout: Modular & Feature-Based Structure

- Core Principle
  - Use a **modular, feature-based structure** to group related assets, scenes, and scripts by the system they belong to.
- Key Directories
  - **res://Entities/**: For characters like Player_Angel/ and EnemyAI/.
  - **res://Systems/**: For **Autoloads/Singletons** (e.g., GameFlow, DataManager).
  - **res://addons/bin/**: The target folder for **compiled GDExtension binaries** (.dll, .so).
- Code Conventions
  - **Mandatory Commenting** on all public methods and complex logic blocks.
  - ~~Use of patterns like **Service Locator** or **Dependency Injection** to keep systems decoupled.~~

Task Delegation: Motivation-Driven Chunking (MDC)

- MDC Strategy
  - A **High-Autonomy, Low-Dependency Model** that prioritizes volunteer motivation and skill-building over traditional rigid deadlines.
- Task Delegation
  1. **Match Interest:** Delegate tasks based on a volunteer's **passion or desired learning path** (e.g., if they want to learn C++, give them a non-critical GDExtension task).
  2. **Low Dependency:** Tasks must be **Atomic, Self-Contained Chunks** (target 4-8 hours of work).
- Outcome
  - This strategy is designed to minimize the risk of **bottlenecks and attrition**, ensuring continuous progress even with inconsistent availability.

Technical Roadmap: Phases 1, 2, and 3

- Phase 1: Vertical Slice
  - **Milestone:** Core combat loop functional; Godot configuration validated.
  - **Focus:** Player movement, basic enemy AI, **Dual-State Transition** prototype, and Lust Reprieve safe zone.
- Phase 2: Alpha
  - **Milestone:** All systems integrated; First two boss fights completed.
  - **Focus:** Finalizing Rogue-lite Persistence, ~~**Soul Power Calculation** (T2.1)~~, and Boss AI Framework (T2.5).
- Phase 3: Beta
  - **Milestone:** Feature complete; Full polish.
  - **Focus:** Final boss logic (**Melancholy Choice Logic**), Act 3 ability injection, Performance Optimization Pass, and Full UI/UX implementation.

Risks & Contingencies for a Volunteer Team

- Risk 1: Volunteer Attrition/Burnout
  - **Mitigation:** Enforce a strict **"No Guilt" Culture** (life comes first) and actively promote the MDC strategy.
  - **Contingency: "Stale Task" Protocol**—if a task is untouched for 3 weeks without an update, it is automatically moved back to "Help Wanted" to free up work.
- Risk 2: Knowledge Silos/Bus Factor
  - **Mitigation:** All complex systems (e.g., Soul Power Calculation) must be **co-authored or reviewed** by at least (one?, two?) distinct individuals.
  - **Contingency:** Maintain a **System Ownership Log** to track who is familiar with each critical system.
- Risk 3: Scope Creep
  - **Mitigation:** All new feature ideas must be approved against the **Core Technical Goals**.

Immediate Next Steps: Finalizing Technical Blueprints

- Goal
  - The team must define the **specific technical artifacts** requested in the TDD blueprint expansion before development starts in Phase 1.
- Core Blueprints Needed

  These are the critical items still requiring concrete definitions:
  - **Player:** Player State Machine Diagram (3.1).
  - **AI:** Base Enemy AI FSM Diagram (5.1).
  - **Data:** MemoryFragment Data Schema (3.2).
  - ~~**Formula:** Celestial_Might Calculation Formula (3.3).~~
  - **Utilities:** Persistence/Save Protocol (6.3).